MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, LOW-POWER,
MULTIPLE NOR GATES, MONOLITHIC SILICON

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic, silicon, TTL, low-power, positive NOR logic gating microcircuits. Three product assurance classes and a choice of case outline/lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The complete part number shall be as shown in the following example:

M38510	/027	01	B	A	A
Military designator	Detail specification	Device type (1.2.1)	Device class (1.2.2)	Case outline (1.2.3)	Lead finish (3.3)

1.2.1 Device type. The device type shall be as shown in the following:

Device type	Circuit
01	Quadruple 2-input positive NOR gate

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	MIL-M-38510, appendix C case outline
A	F-1 (14 pin, 1/4" x 1/4", flat-pack)
B	F-3 (14-pin, 1/4" x 1/8", flat-pack)
C	D-1 (14-pin, 1/4" x 3/4", dual-in-line pack)
D	F-2 (14-pin, 1/4" x 3/8", flat-pack)

1.3 Absolute maximum ratings.

Supply voltage range	0 Vdc to 7.0 Vdc
Input voltage range	0 Vdc to 5.5 Vdc
Storage temperature range	-65°C to 150°C
Maximum power dissipation per gate, P_D	4.2 mWdc 1/
Lead temperature (soldering 10 seconds)	300°C
Thermal resistance, junction to case	θ_{JC} = {0.09°C/mW for flat pack; 0.08°C/mW for dual-in-line pack}
Junction temperature	T_J = 175°C

1/ Must withstand the added P_D due to short circuit condition (e.g., I_{OS}) at one output for 5 seconds duration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:58Z
Check the source to verify that this is the current version before use.

1.4 Recommended operating conditions.

Supply voltage	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage	2.0 Vdc
Maximum low level input voltage	0.7 Vdc
Normalized fanout (each output) 2/	10 maximum
Ambient operating temperature range	-55°C to 125°C

2. APPLICABLE DOCUMENT

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuit. The schematic circuit shall be as specified on figure 3, or equivalent.

3.2.4 Case outline.

3.2.4.1 Case outline A. Case outline A shall be in accordance with F-1 of appendix C to MIL-M-38510.

3.2.4.2 Case outline B. Case outline B shall be in accordance with F-3 of appendix C to MIL-M-38510.

3.2.4.3 Case outline C. Case outline C shall be in accordance with D-1 of appendix C to MIL-M-38510.

3.2.4.4 Case outline D. Case outline D shall be in accordance with F-2 of appendix C to MIL-M-38510.

3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510.

2/ Device will fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:58Z
Check the source to verify that this is the current version before use.

3.4 Electrical performance characteristics. The electrical performance characteristics are specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions	Limits		Units
			Min	Max	
High-level output voltage (except open collector output circuits)	V_{OH}	V_{CC} = 4.5 V, V_{IN} = 0.7 V $-I_{OH}$ = -100 µA 2/	2.4	---	Volts
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 2.0 mA V_{IN} = 2.0 V for all inputs of gate under test 1/	---	0.3	Volts
High-level input current	I_{IH_1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V 1/	---	10	µA
High-level input current	I_{IH_2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V 1/	---	100	µA
Low-level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.3 V 2/	-60	-180	µA
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V 2/ 3/	-3.0	-15	mA
High-level supply current per gate	I_{CCH}	V_{CC} = 5.5 V 2/ V_{IN} = 0 V	---	0.4	mA
Low-level supply current per gate	I_{CCL}	V_{CC} = 5.5 V 1/ V_{IN} = 5.5 V	---	0.65	mA
Propagation delay time, high-to-low-level	t_{PHL}	C_L = 50 pF R_L = 4 kΩ	3	80	ns
Propagation delay time, low-to-high-level	t_{PLH}	C_L = 50 pF R_L = 4 kΩ	3	80	ns

1/ All unspecified inputs grounded.
2/ All unspecified inputs at 5.5 V.
3/ Not more than one output should be shorted at a time.

3.5 Rebonding. Rebonding shall be in accordance with MIL-M-38510.

3.6 Electrical test requirements. Electrical test requirements shall be as specified in table III for the applicable device type and device class. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification and quality conformance, by device class are specified in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:58Z
Check the source to verify that this is the current version before use.

Device type 01

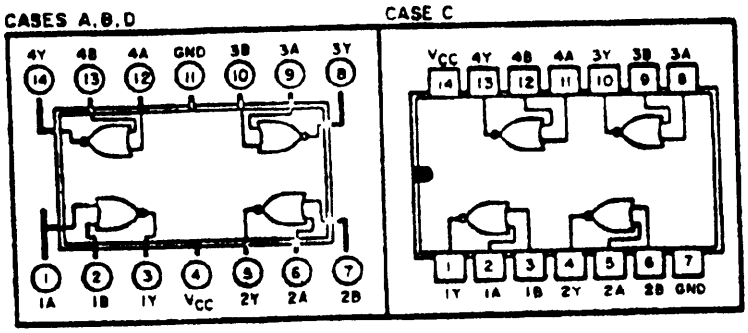


FIGURE 1. Logic diagram and terminal connections (top view).

Device type 01

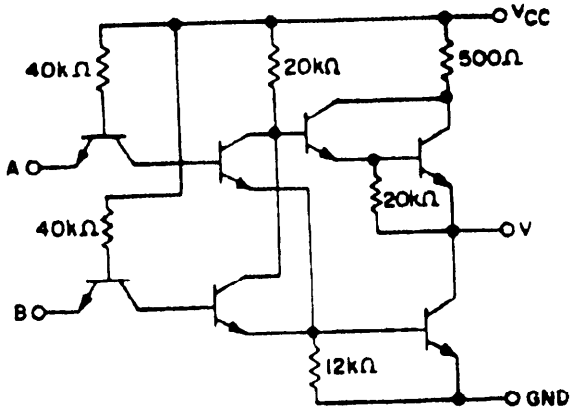
Truth table each gate		
Input		Output
A	B	Y
H	X	L
X	H	L
L	L	H

X = Irrelevant

Positive logic: $Y = \overline{A + B}$

FIGURE 2. Truth tables and logic equations.

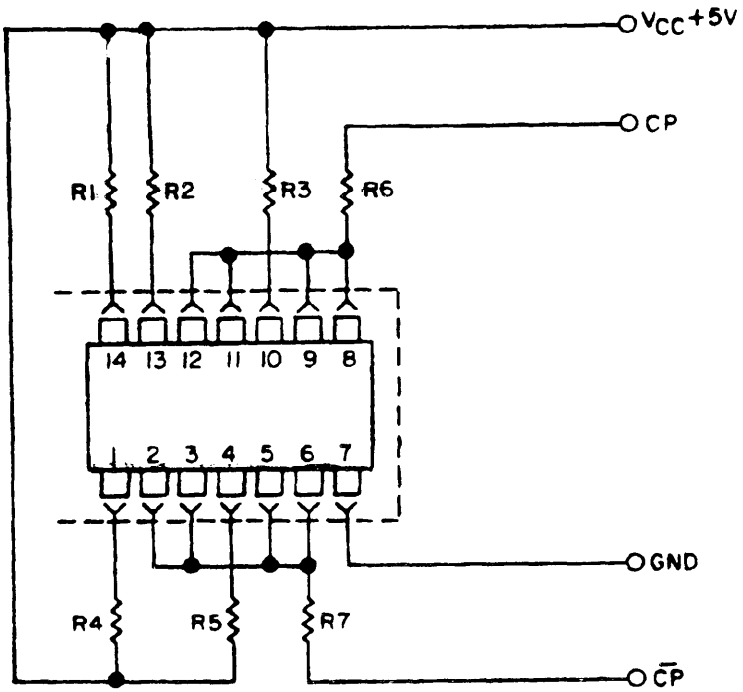
Device type 01



NOTE: Component values shown are nominal.

FIGURE 3. Schematic circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:58Z
Check the source to verify that this is the current version before use.

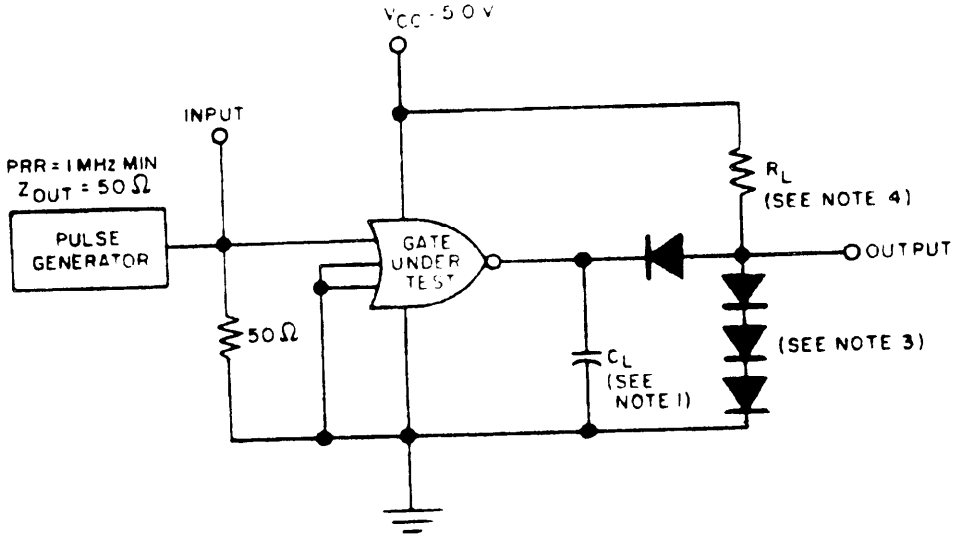


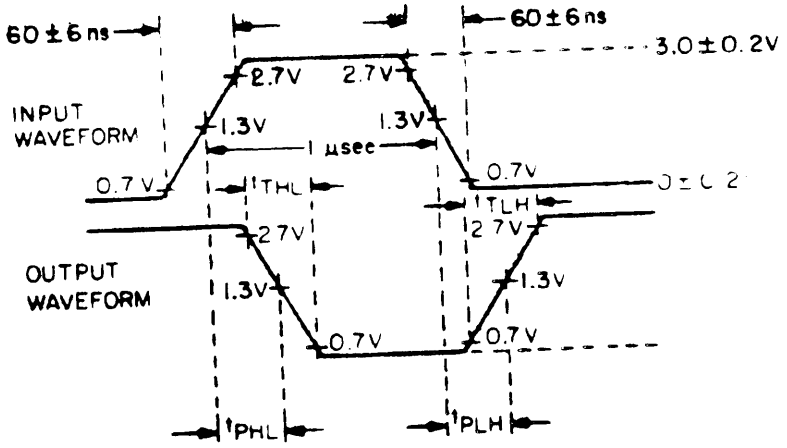
R1, R6 & R7 = 26.1Ω
R2, R5 = 2.43 KΩ
CP & $\overline{CP}$ = 100 KHZ, 0-3V, 50 W

NOTES:
1. One input of each gate shall be driven. Remaining gate inputs shall be connected to GND.
2. V_{CC} = 5.0V.
3. If parallel excitation is used, clock frequency shall be 100KHz with minimum V_{IN} = 3V and minimum duty cycle of 50%.

FIGURE 4. Burn-in and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:58Z
Check the source to verify that this is the current version before use.





NOTES:
1. C_L = 50pF minimum including scope probe, wiring, and stray capacitance, without package in test fixture.
2. Voltage measurements are to be made with respect to network ground terminal.
3. All diodes are 1N3064 or equivalent.
4. R_L = 4kΩ ± 5%.

FIGURE 5. Switching time test circuit for device type 01.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:58Z
Check the source to verify that this is the current version before use.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirement	Subgroups (see table III)		
	Class A devices	Class B devices	Class C devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1	None
Final electrical test parameters (method 5004)	1*, 2, 3, 9	1*, 2, 3, 9	1
Group A test requirements (method 5005)	1, 2, 3, 9, 10, 11	1, 2, 3, 9	1, 2, 3, 9
Groups B and C end point electrical parameters (method 5005)	1, 2, 3	1, 2, 3	1
Additional electrical subgroups for group C periodic inspections	None	10, 11	None

* PDA applies to subgroup 1 (see 4.3(h)).

3.7 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein. At the option of the manufacturer, the following marking may be omitted from the body of the microcircuit, but shall be retained on the initial container:

Country of origin.

4. PRODUCT ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and method 5005 of MIL-STD-883, except as modified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, and C inspections (see 4.4.1, 4.4.2, and 4.4.3). After qualification of one or more electrically and structurally similar types with a single lead finish, other lead finishes of the same case outline may be qualified by submitting a single type in the qualified case outline to the group B, subgroup 3 test and the group C, subgroups 1, 3, and 4 tests.

4.3 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

(a) Test samples for the group B bond strength test specified in method 5005 of MIL-STD-883 may, at the manufacturer's option, be randomly selected immediately following the internal visual (precap) inspection and prior to sealing (see 4.4.2(b)).
(b) Temperature cycling (method 1010 of MIL-STD-883).
    (1) Omit seal test as post-test measurements.
(c) Thermal shock (method 1011 of MIL-STD-883), when substituted for temperature cycling.
    (1) Omit seal test as post-test measurements.
(d) Burn-in test (method 1015 of MIL-STD-883).
    (1) Test condition D or E, using the circuit shown on figure 4, or equivalent.
    (2) T_A = 125°C minimum.
(e) Reverse bias burn-in and interim electrical test in accordance with 3.1.10 of method 5004 of MIL-STD-883 may be omitted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:58Z
Check the source to verify that this is the current version before use.

(f) Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.
(g) External visual inspection shall not include measurement of case and lead dimensions.
(h) Percent defective allowable (PDA) - The PDA is specified as 5 percent for class A devices and 10 percent for class B devices based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510.

4.4.1 Group A inspection. Group A inspection shall consist of the test subgroups and LTPD values shown in table I of method 5005 of MIL-STD-883 and as follows:

(a) Tests shall be as specified in table II.
(b) Subgroups 4, 5, 6, 7, and 8 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall consist of the test subgroups and LTPD values shown in table II of method 5005 of MIL-STD-883 and as follows:

(a) End point electrical parameters shall be as specified in table II.
(b) Bond strength test may be conducted on samples collected prior to sealing (see 4.3(a)).

4.4.3 Group C inspection. Group C inspection shall consist of the test subgroups and LTPD values shown in table III of method 5005 of MIL-STD-883 and as follows:

(a) End point electrical parameters shall be as specified in table II.
(b) Subgroups 7 and 8 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.
(c) Lead bend in only one direction is required for initial conditioning prior to moisture resistance and salt atmosphere tests.
(d) High temperature storage test (method 1008 of MIL-STD-883) conditions:
  (1) Temperature: 150 ±10°C.
  (2) Duration: 1,000 hours, except as otherwise permitted by appendix B of MIL-M-38510.
(e) Operating life test (method 1005 of MIL-STD-883) conditions:
  (1) Test condition D or E, using the circuit shown on figure 4, or equivalent.
  (2) T_A = 125°C, minimum.
  (3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510.
(f) Omit steady state reverse bias test.

4.5 Methods of examination and test. Methods of examination and test shall be as specified in the appropriate tables and as follows.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:58Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas terminal	Test limits		
			Case C	2	3	1	14	4	5	6	10	8	9	7	11	12	13				
			Test No.	1A	1B	1Y	VCC	2Y	2A	2B	3Y	3A	3B	GND	4A	4B	4Y		Min	Max	Unit
1 T_A = 25°C	V_{OL}	3007	1	2.0 V	GND	2.0 mA	4.5 V		GND	GND		GND	GND	GND	GND	GND		1Y		0.3	V
			2	GND	2.0 V	2.0 mA												1Y			
			3		GND			2.0 mA	2.0 V									2Y			
			4					2.0 mA	GND	2.0 V								2Y			
			5							GND	2.0 mA	2.0 V						3Y			
			6								2.0 mA	GND	2.0 V					3Y			
			7										GND		2.0 V		2.0 mA	4Y			
			8										GND		GND	2.0 V	2.0 mA	4Y			
	V_{OH}	3006	9	0.7 V	0.7 V	-100 µA		-100 µA	5.5 V	5.5 V		5.5 V	5.5 V		5.5 V	5.5 V		1Y	2.4		
			10	5.5 V	5.5 V				0.7 V	0.7 V								2Y			
			11						5.5 V	5.5 V	-100 µA	0.7 V	0.7 V					3Y			
			12						5.5 V	5.5 V		5.5 V	5.5 V		0.7 V	0.7 V	-100 µA	4Y			
	I_{OS}	3011	13	GND	GND	GND	5.5 V	GND	GND	GND								1Y	-3.0	-15	mA
			14								GND	GND	GND					2Y			
			15															3Y			
			16												GND	GND	GND	4Y			
	I_{IH_1}	3010	17	2.4 V	GND	2.4 V			GND	GND		GND	GND		GND	GND		1A		10	µA
			18	GND	2.4 V	GND			2.4 V									1B			
			19		GND				GND	2.4 V								2A			
			20							GND		2.4 V						2B			
			21									GND	2.4 V					3A			
			22										GND					3B			
			23												2.4 V			4A			
			24												GND	2.4 V		4B			
	I_{IH_2}	3010	25	5.5 V	5.5 V											GND		1A		100	
			26	GND	GND				5.5 V									1B			
			27						GND	5.5 V								2A			
			28							GND		5.5 V						2B			
			29									GND	5.5 V					3A			
			30										GND					3B			
			31												5.5 V			4A			
			32												GND	5.5 V		4B			
	I_{IL}	3009	33	0.3 V	5.5 V				5.5 V	5.5 V		5.5 V	5.5 V		5.5 V			1A	-0.06	-0.18	mA
			34	5.5 V	0.3 V				0.3 V									1B			
			35		5.5 V				5.5 V	0.3 V								2A			
			36							5.5 V		0.3 V						2B			
			37									5.5 V	0.3 V					3A			
			38										5.5 V		0.3 V			3B			
			39												5.5 V	0.3 V		4A			
			40															4B			
	I_{CCH}	3005	41	GND	GND				GND	GND		GND	GND		GND	GND		VCC		1.6	mA
	I_{CCL}	3005	42	5.5 V	GND				5.5 V	GND		5.5 V	GND		5.5 V	GND		VCC		2.6	mA

2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125°C.
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55°C.

Source: https://assist.dla.mil -- Downloaded: 20260-21T12:58Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01. - Continued

Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A,B,D Case C Test No.	1 2 1A	2 3 1B	3 1 1Y	4 14 VCC	5 4 2Y	6 5 2A	7 6 2B	8 7 GND	9 8 3A	10 9 3B	11 7 GND	12 11 4A	13 12 4B	14 13 4Y	Meas. terminal	Test limits Min	Test limits Max	Unit
9 T_A = 25°C	t_{PHL}	3003 (Fig. 7)	43	IN	GND	OUT	5.0V	OUT	IN	GND		IN	GND		IN	GND	OUT	1A to 1Y	3	46	ns
			44															2A to 2Y			
			45															3A to 3Y			
			46															4A to 4Y			
	t_{PLH}		47	IN	GND	OUT		OUT	IN	GND		IN	GND		IN	GND	OUT	1A to 1Y	3	46	
			48															2A to 2Y			
			49															3A to 3Y			
			50															4A to 4Y			
10 T_A = 125°C	t_{PHL}		51	IN	GND	OUT		OUT	IN	GND		IN	GND		IN	GND	OUT	1A to 1Y	3	80	
			52															2A to 2Y			
			53															3A to 3Y			
			54															4A to 4Y			
	t_{PLH}		55	IN	GND	OUT		OUT	IN	GND		IN	GND		IN	GND	OUT	1A to 1Y	3	80	
			56															2A to 2Y			
			57															3A to 3Y			
			58															4A to 4Y			

11 Same tests, terminal conditions and limits as for subgroup 10, except T_A = -55°C.

Source: https://assist.dla.mil -- Downloaded: 20260-21T12:58Z
Check the source to verify that this is the current version before use.

Preparing activity:
Air Force - 17

Agent:
DSA - ES

(Project 5962-F052)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:58Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER

2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:58Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE AIR FORCE

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

RADC (RBRC)
Griffiss AFB, NY 13440

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:58Z
Check the source to verify that this is the current version before use.